Exercise : Cyclic Code

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Sol cyclic code for (7,4) we can use $g(x) = x^3 + x + 1$

$= 1011$

0101	1001	0010	1110	0000

shift → 0101000 | 1001000 | 0010000 | 1110000 | 0000000

divide by 1011 →

1011) 0101000 → 1011, 100

1011) 1001000 → 1011, 1000, 1011, 110

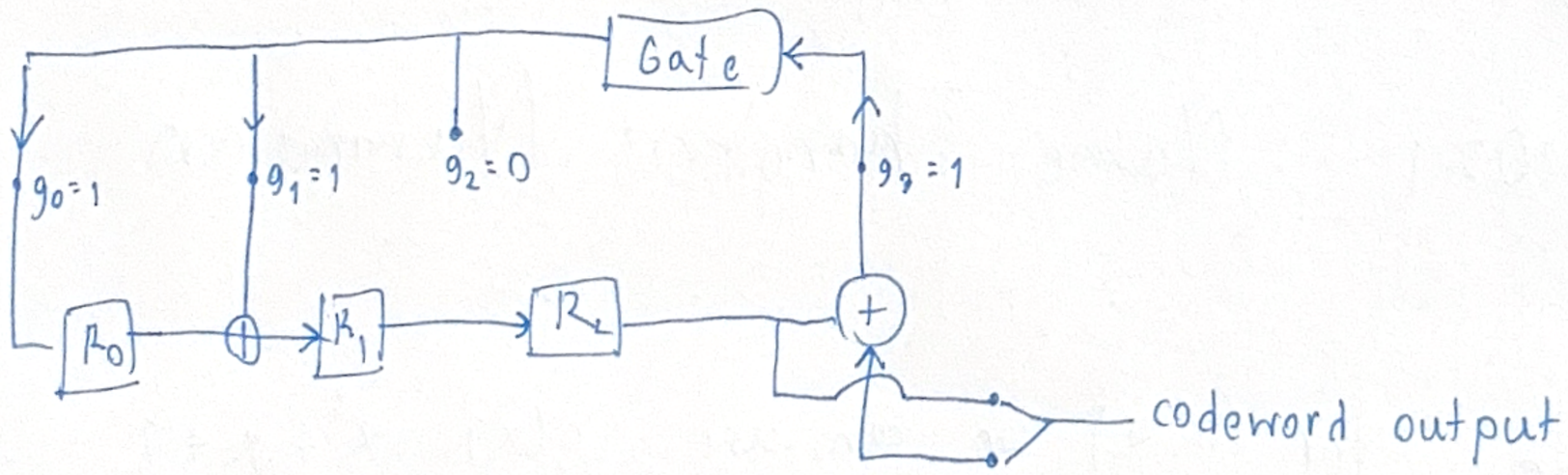
1011) 0010000 → 1011, 110

1011) 1110000 → 1011, 1010, 1011, 100

1011) 0000000 → 000

codeword → 0101100 | 1001110 | 0010110 | 1110100 | 0000000

Hardware Implementation



message

For the first 4 clock cycle, the swith transmit raw message while LFSR calculate the remainder, for the final 3, Gate disable feedback loop and the switch toggles to flush the stored parity to the output.